Abdullah Ibne Tayeb Tamur

Chattogram, Bangladesh | +880 1329463330 | abdullahibnetayebtamur@gmail.com
Portfolio | GitHub | Google Scholar | ORCID | LinkedIn

RESEARCH PROFILE

Research Interests: Medical AI, Explainable AI (XAI), Multimodal Clinical AI, Medical Image Analysis, Federated Learning, Data-Efficient Medical AI, and Clinical Decision Support Systems.

First author of three peer-reviewed conference papers and co-author of two additional peer-reviewed conference papers, with further medical AI manuscripts currently under review. Recent work includes *CHD-Net*, an explainable pediatric congenital heart disease classification framework accepted at IEEE QPAIN 2026; *CMA-Fusion*, a cross-modality attention framework for Alzheimer's disease detection accepted at ICCA 2026; and *HemX*, an explainable hematological triage framework evaluated on 13,823 patient records and currently under review at *Biomedical Signal Processing and Control*. Research focuses on clinically useful, interpretable, and validated AI systems for medical imaging, clinical triage, multimodal healthcare data, and decision support.

EDUCATION

International Islamic University Chittagong (IIUC) *Bangladesh*
Bachelor of Science in Computer Science and Engineering Jan 2022 – Jun 2026
CGPA: 3.416 / 4.00 *Last 4 semesters GPA: 3.71 / 4.00*

RESEARCH EXPERIENCE

Elite Research Lab LLC *Remote*
Research Intern Jun 2026 – Present

Conducting applied research on Explainable AI, clinical diagnostic systems, Retrieval-Augmented Generation, and dual-mode LLM frameworks under formal NDA. Developing interpretable machine learning pipelines for healthcare applications with attention to clinical usability, privacy, and deployment constraints.

Computational Intelligence and Society (CIS) Lab, IIUC *Chattogram, Bangladesh*
Research Mentor Jan 2024 – Present

Mentoring undergraduate researchers through literature review, experimental design, model development, and manuscript preparation. Guided multiple project teams toward peer-reviewed conference submissions; supported implementation in PyTorch, computer vision architectures, attention mechanisms, and XAI tools.

PUBLICATIONS

Under Review

[1] **Tamur, A. I. T.** et al. (**First Author**), *HemX: Hierarchical XAI for Hematological Triage.* Under review at **Biomedical Signal Processing and Control** (Elsevier).
Explainable hematological triage framework evaluated on 13,823 patient records. Integrates global SHAP, patient-level LIME, and the Hematological Heterogeneity Index (HHI) as an engineered biomarker. Evaluated using TRIPOD+AI-guided reporting and stratified 5-fold cross-validation.

[2] **Tamur, A. I. T.** et al. (**Second Author**), *Fed-NeuroTriage: Federated Distillation Architecture for Psychiatric NLP.* Under review at **JSCDM**.
Privacy-aware federated learning framework for psychiatric NLP triage using MIMIC-IV discharge summaries. Addresses severe class imbalance through knowledge distillation and inverse-frequency weighting.

Peer-Reviewed Publications and Accepted Conference Papers

[3] **Tamur, A. I. T.** et al. (**First Author**), *Cross Modality Attention for Leakage-Free Alzheimer's Disease Detection Using 3D MRI and Clinical Biomarkers.* **ICCA 2026.** Accepted for presentation.
Cross-modality attention framework for Alzheimer's disease detection using 3D MRI and clinical biomarkers. The study focuses on leakage-free experimental design, multimodal feature integration, and clinically relevant decision-support modeling for neurodegenerative disease detection.

[4] **Tamur, A. I. T.** et al. (**First Author**), *CHD-Net: An Explainable Deep Learning Framework for Pediatric Congenital Heart Disease Classification.* **IEEE QPAIN 2026. Best Presenter Award.**
Explainable ensemble framework for pediatric congenital heart disease classification from echocardiography images. Achieved 99.15% accuracy and 99.99% AUC on 4,140 echocardiography images derived from 828 source scans using stratified 5-fold cross-validation. SHAP GradientExplainer was applied for model interpretation and anatomical localisation of IAS, IVS, and ductus arteriosus regions.
DOI: 10.1109/QPAIN69676.2026.11545695

[5] **Tamur, A. I. T.** et al. (**First Author**), *Dual-Stream FreqViT: FFT and Vision Transformer Fusion for Robust Plant Disease Detection.* **IEEE QPAIN 2026.**
Dual-stream architecture combining a spatial CNN branch, 2D FFT spectral branch, and Transformer encoder fusion. Achieved 97.75% accuracy under challenging field-lighting conditions.
DOI: 10.1109/QPAIN69676.2026.11546204

[6] **Tamur, A. I. T.** et al. (**Third Author**), *FUSE SENet: Fusion with SE Attention Network for Robust Skin Lesion Classification.* **IEEE ICCIT 2025.**
Fusion-based feature learning with squeeze-and-excitation (SE) attention for robust skin lesion classification.
DOI: 10.1109/ICCIT68739.2025.11491158

[7] **Tamur, A. I. T.** et al. (**Fifth Author & Research Mentor**), *DiaXAI-Stack: An Explainable AI Framework for Diabetes Prediction Using Leakage-Free Stacking Ensemble Learning with Dual Threshold Optimization.* **ICCA 2026.** Accepted for presentation.
Explainable AI framework for diabetes prediction using leakage-free stacking ensemble learning and dual threshold optimization. Contributed as research mentor by supporting dataset preprocessing, SHAP/LIME integration for clinical transparency, experimental design, and manuscript development.

AWARDS & HONORS

Best Presenter Award — IEEE QPAIN Conference 2026, awarded for the presentation of the *CHD-Net* clinical AI framework.

AgentX AI Prompting Competition — Ranked **6th among 700+ university participants** at the IIUC level and participated among a large national pool of university competitors in Bangladesh.

DEPLOYMENTS & ENGINEERING PROJECTS

HemoForge – Explainable Hematology Engineering Platform: Full-stack CBC analysis platform integrating a 9-class HistGradientBoosting model, SHAP explanations, out-of-distribution (OOD) assessment, selective review indicators, and a 10-node LangGraph workflow with LangSmith tracing. HemoParse supports human-verified PDF/image CBC extraction using PyMuPDF, pdfplumber, Tesseract OCR, and Groq LLM fallback; FastAPI, Next.js/TypeScript, and SQLite support the API, interface, case archive, and audit trail. The platform is being extended with *HemoAssist*, a Doctor/Patient LLM assistant for plain-language report explanation, model-result interpretation, and controlled scholarly-literature retrieval.

HemaTriage: Clinical web application based on the HemX architecture for CPU-only hematological triage inference in resource-constrained healthcare settings.

Customer Support Agent [GitHub]: LangGraph + Groq (Llama 3.3 70B) agent with triage classification, six domain-specific tools, input/output guardrails, human-in-the-loop approval, short-term memory, LangSmith observability, and SQLite persistence.

SpectraLeaf Analytics [Live]: Cloud-deployed plant disease detection application combining spatial CNN features with FFT-based spectral representations for robust field-condition inference.

Wolf Scholar [GitHub]: Multimodal RAG-based research assistant using LangChain, FAISS, and Groq for multi-document analysis and medical vision research workflows.

Autonomous FPV Drone: Arduino and GPS-based autonomous navigation system with 1.5 km operational range and real-time video transmission.

LEADERSHIP & SERVICE

Assistant Social Welfare Secretary, IIUC Computer Club — Organised mental health awareness programmes and coordinated peer academic support initiatives for the university computing community.

Volunteer – 2024 Flood Relief Operations — Collaborated with As-Sadaqa and the Bangladesh Red Crescent Society to prepare and distribute emergency relief packages for flood-affected communities across Chittagong Division.

Home Tutor *2020 – Present*
Teaching secondary and higher-secondary students in Biology, English, ICT, and Chemistry; providing personalised academic support and exam preparation guidance alongside undergraduate studies and research.

TECHNICAL SKILLS & METHODOLOGIES

Research Methods: Explainable AI, SHAP, LIME, Federated Learning, LangGraph Agents, Conformal Prediction, Uncertainty Quantification, Clinical Model Validation, Stratified K-fold Cross-Validation, TRIPOD+AI, PROBAST+AI

Languages: Python, C++, SQL, HTML/CSS, JavaScript

Frameworks & Tools: PyTorch, TensorFlow, Keras, OpenCV, LangChain, LangGraph, Flask, FAISS, Git, LaTeX, Jupyter Notebook